

"Smart traffic light control"

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Introduction

Modern cities often face the growing problem of traffic congestion. The increase in the number of vehicles and high traffic density leads to increased travel time, pollutant emissions and an increased likelihood of accidents [2]. To address this problem, intelligent transport systems are being developed, including smart traffic management using sensors and intelligence. For example, within the framework of the Astana smart city project, it is planned to implement both a solution and a network of IoT sensors for digital traffic modeling and increasing the efficiency of transport flows [5].

However, existing systems are not able to effectively adapt to traffic dynamics [2]. This causes problems such as downtime at intersections, which in turn leads to additional CO₂ emissions and loss of time [3]. In this regard, the "Smart Traffic light Control" research aims to develop adaptive traffic light control methods that will minimize congestion and environmental costs.

Research questions:

1. How intelligent traffic light control algorithms can reduce downtime and emissions in urban traffic conditions?
2. What technologies and data models are best used for real-time traffic control?
3. What are the technical and organizational barriers to the implementation of such systems in the existing infrastructure?

Relevance and important of the research:

The costs of traffic jams and air pollution have a major socio-economic impact. The transport sector produces approximately 25% of global CO₂ emissions, with more than 70% of these coming from road transport [3]. It is estimated that installing smart traffic lights can significantly reduce these figures. Emissions can be reduced by 32-40%, and waiting time in traffic jams can be reduced by 53-95% [3]. In Astana's example, such solutions promise not only to reduce congestion but also to improve safety[5].

Literature review

Key concepts: Intelligent traffic control is based on a combination of classic traffic lights with sensors and machine learning algorithms for adaptive adjustment of the phases of light cycles. In such systems, video cameras, inductive loops, radars, and other sensors are widely used to collect data on the current density and speed of traffic[1,2,7]. AI-based algorithms, such as convolutional neural networks, analyze this data and predict traffic congestion and dynamically change the duration of traffic lights in each direction [1,2,7]. Several studies have proposed hybrid approaches, which, in addition to optimizing traffic, also regulate street lighting to save energy.

Main studies: Modern literature describes numerous prototypes and simulation models of smart traffic lights. For example, Zraqou and etc. (2022) they propose using a hybrid system of IoT sensors and information systems for load forecasting[2]. Chinedu and etc. (2025) demonstrate a pilot control system based on computer vision and microcontrollers. The camera counts the number of cars and transmits it to the Arduino board, which adjusts the signal duration in real time [1]. These studies show us the effectiveness of the hybrid approach in reducing traffic jam time and optimizing energy consumption.

Key debates and controversies: Various plans and strategies are being discussed in the Its area. One of the directions focuses on communications, transport infrastructure or transport, it promises coordination between traffic lights and cars. However, the authors point out that the practical implementation of such communications is limited by the need to equip older vehicles with special equipment, and such solutions are currently difficult to scale [3]. Another issue is security and privacy, as the proliferation of cameras and the collection of personal data has raised some controversy. There are also debates about the priority of public transport; smart systems can give green signals to buses and trams, improving the environment [6].

Gaps in existing knowledge: Despite a large number of studies, many simplify the conditions aimed at megacities. As Santos and others (2023) note, the results of classical models are difficult to transfer to small cities due to different driving styles [3]. Ready (2015) also points out that existing prototypes require improvements to both hardware and software for their stable operation in real conditions[9]. Thus, there is a need for research that uses realistic traffic models and does not require fully equipping vehicles with new electronics. Another gap is the lack of data on the effect of smart traffic lights in the long term and in different situations. Our proposal aims to fill these missing problems by developing a control algorithm tested on an urban traffic model.

Research design and methods

The project is based on a quantitative approach using computer modeling. The main goal is to simulate the environment as, for example, a task in SUMO for testing an algorithm. Proposed work plan:

1. Firstly, traffic simulation involves developing a model of the intersection. An approach similar to that of Sentos et al. (2023) will be used, with realistic kinematics and precise emission calculations. Here various traffic densities and flow distributions will be included
2. Secondly, this is the processing of the data, which includes the implementation of a vehicle detection algorithm based on machine learning. For example, Haar cascade or CNN technology will be applied to frames from a camera recording an intersection, similar to the Chinedu et al. (2025). The calculated number of vehicles will be transmitted to the signal control controller.
3. Thirdly, it is signal analysis, that is, the development of an algorithm for dynamically determining the signal duration, which in turn minimizes the total waiting time and emission. It is planned to use the Roy method or RL algorithms.
4. and the very last one is the validation of our results, that is, a comparison of the performance of the intelligent system with the existing one. The assessment will be based on metrics such as average waiting time, throughput, and total CO₂ emissions.

Open statistical data or empirical flow curves will be used in the modeling. Where possible, public video surveillance recordings or similar datasets will be used. Large smart projects plan to install hundreds of IoT sensors and cameras [5]. Therefore, in the future, real city data may be available for model verification.

Practical aspects and limitations: The main difficulties associated with the availability of traffic data and computational resources for modeling. Accurate emission measurement requires detailed kinematics, which complicates calculations. In addition, the implementation of a real prototype requires significant investment in equipment, but a simulated experiment will help justify the effectiveness of the approach. It is also necessary to take into account the possible effect of unpredictable factors such as weather conditions or accidents, which are taken into account conditionally in the model. These limitations will be reduced by conducting simulations with different scenarios and their analysis.

Contribution to science

Practical implications: The system is expected to offer significant improvements in traffic management based on existing models such as those described by Santos and others (2023), their example demonstrated a reduction in traffic jams and emissions. Installing a smart traffic light at one intersection could reduce CO₂ emissions by up to 40% and speed up traffic by up to 101% [3]. Such solutions will allow city authorities to reduce time costs and reduce pollution. This work could form the basis for making decisions about installing computer vision systems and IoT controllers at intersections.

Theoretical implications: The study could expand our understanding of how artificial intelligence algorithms work in transport modeling. The novelty lies in the realistic simulation and assessment of the impact of management on emission indicators. This will contribute to the theory of such systems by showing how to balance between the priority of public transport and overall capacity. This method can only serve as a basis for further research in the field of hybrid systems and provide grounds for the development of new models.

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Research schedule

Phase	Goals	Deadline
Preparatory	Literature collection and analysis, refinement of objectives and methodology	November 2025
Model development	Construction of a simulation model of intersections and flows	December 2025
Algorithm implementation	Development and integration of detection and control algorithms	January 2026
Experimentation	Running simulations, collecting results, analyzing effectiveness	February 2026
Report writing	Presentation of results, conclusions, and recommendations	March 2026
Final submission	Revision, preparation of the final version of the proposal	April 2026